

数学物理方法 II 第三次作业

Due date: 2024/04/xx

1. 将下列二阶偏微分方程化为标准形式。

$$(1) \quad \frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 u}{\partial x \partial y} + 5 \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0;$$

$$(2) \quad \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial y^2} + \frac{1}{2} \frac{\partial u}{\partial y} = 0;$$

$$(3) \quad \frac{\partial^2 u}{\partial x^2} - 2 \cos x \frac{\partial^2 u}{\partial x \partial y} - (3 + \sin^2 x) \frac{\partial^2 u}{\partial y^2} - y - \frac{\partial u}{\partial y} = 0$$

2. 将以下的方程分离变量：

$$(1) \quad a_1(x) \frac{\partial^2 u}{\partial x^2} + b_1(y) \frac{\partial^2 u}{\partial y^2} + a_2(x) \frac{\partial u}{\partial x} + b_2(y) \frac{\partial u}{\partial y} = 0$$

$$(2) \quad \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 0$$

$$(3) \quad \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = 0$$

3. 使用分离变量法求解如下波动方程的问题：

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, (0 < x < l, t > 0) \\ \frac{\partial u}{\partial x} \Big|_{x=0} = 0, \frac{\partial u}{\partial x} \Big|_{x=l} = 0 \\ u|_{t=0} = e^{-x^2}, \frac{\partial u}{\partial t} \Big|_{t=0} = 2axe^{-x^2} \end{cases}$$

其中参数自己设定，使用 Matlab 等数学软件画出分离变量法的数值结果的图像（比如 n 取 100 时的解的情况），任取一些 t 显示解的演化。

4.

求解细杆的导热问题，杆长为 l ，两端 ($x=0$ 及 $x=l$) 均保持 0 度，初始温度分布 $u|_{t=0} = bx(l-x)/l^2$ 。

5. 求解下述三维波动方程：

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ u|_{t=0} = \begin{cases} u_0, x^2 + y^2 + z^2 < R^2 \\ 0, x^2 + y^2 + z^2 > R^2 \end{cases}, \frac{\partial u}{\partial t} \Big|_{t=0} = 0 \end{cases}$$

6. 解环内泊松方程的定解问题：

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6(x+y)^2, 0 < a^2 < x^2 + y^2 < b^2 < \infty \\ u|_{x^2+y^2=a^2} = 1, \frac{\partial u}{\partial n} \Big|_{x^2+y^2=b^2} = 0 \end{cases}$$

其中， n 为边界 $x^2 + y^2 = b^2$ 的外法向。

7. 对比第二次作业中半无界弦振动问题的初值问题，现改为定义在 $[0, \ell]$ 上的有界弦，求解振动情况。

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}, & 0 < x < \ell, t > 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 < x < \ell \\ u_x(0, t) = 0 & t > 0 \\ u(\ell, t) = 0 & t > 0 \end{cases}$$

(1) 设初始位置为 $\sin(x)$ ，初始速度为 0，求解 u ，并做图讨论 $t = \frac{n\pi}{3a}$ ($n = 0, 1, 2, 3, 4, 5, 6$) 这 7 个时间点的振动情况。

(2) 将 ℓ 外推到无限大，对比第二次作业中半无界弦振动问题的结果并讨论。